Cloud-Based ML Model Training

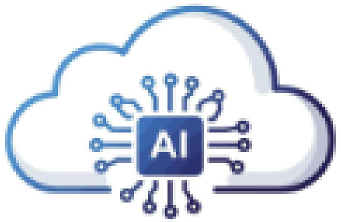
Cloud training builds intelligence for real-world AIoT systems

Why Train Models in the Cloud?

- Massive compute power & scalable storage
- Ability to process large historical datasets

Training with Big Data

- Train ML & deep learning models
- Discover patterns, predictions & insights



Cloud Advantages

- Experimentation & rapid iteration
- Elastic scaling for large workloads
- Faster training & optimization

Model Lifecycle Management

- Versioning & testing
- Validation & performance tuning
- Prepare models for deployment

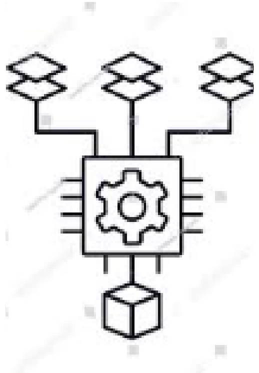
AIoT Integration

- Cloud training + edge inference
- Learn centrally, act locally

Key Steps in Cloud-Based AIoT Model Training

1 Data Collection & Preparation

- IoT devices generate sensor, image & audio data
- Data ingested into cloud storage
- Clean, label & transform for training



2 Model Selection & Training

- Platforms: Sage Maker • Vertex AI • Azure ML
- Train models using scalable GPU compute
- Hyperparameter tuning & validation

3 Model Packaging & Deployment

- Create deployable artifacts (Docker, TFLite, ONNX)
- Secure delivery to edge devices
- Enable offline, low-latency inference

4 Monitoring & Continuous Improvement

- Track performance & edge metrics
- Retrain with fresh data (MLOps lifecycle)

A continuous learning loop keeps AIoT systems accurate and adaptive

Key Benefits

Scalability

- Massive compute & storage on demand
- Train complex models on large datasets

Efficiency

- Managed services handle infrastructure & setup
- Teams focus on model development & innovation

Collaboration

- Shared datasets, models & APIs
- Version control & team-wide access

Edge–Cloud Synergy

- Cloud training for deep intelligence
- Edge inference for real-time decisions

Combine cloud power with edge speed for intelligent AIoT systems

Platforms and Tools

Leading Cloud Platforms

Cloud platforms provide scalable AI training and seamless IoT integration

Amazon Web Services (AWS)

- **SageMaker** for model building & training
- **AWS IoT Core** for device connectivity

Google Cloud Platform (GCP)

- **Vertex AI** for building & scaling **ML models**
- **Integrated AutoML & MLOps** capabilities

Microsoft Azure

- **Azure Machine Learning** for model lifecycle
- **Azure IoT Edge** for device deployment

AI Frameworks

- **TensorFlow**
- **PyTorch**
- Widely supported across all cloud platforms